AFTER A SUITABLE ADMISSIBLE BLOWUP A FINITE TYPE MAP THAT IS AN ISOMORPHISM OVER A DENSE OPEN BECOMES AN OPEN IMMERSION OF THE STRICT TRANSFORM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8961-8974 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8976-8997, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author explicitly flags the lemma as a generalization of lemma-isomorphism-after-blowing-up of the same chapter.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's flattening-after-blowup lemma (lemma-flat-after-blowing-up), the scheme-theoretic-density and effective-Cartier-complement machinery (Divisors on Spaces), and the chapter's lemma-zariski characterizing open immersions under flat finite presentation plus density.

Source reference. Original lemma 087E: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8861–8869. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GUW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8937–8945. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0879: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8649–8657. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 085T: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8997–8997. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 083S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8997–8997. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 086C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8997–8997. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Let $U \subset B$ be an open subspace. Assume*

(1) *B quasi-compact and quasi-separated,*
(2) *U is quasi-compact,*
(3) *f is of finite type*
(4) *$f^{-1}(U) \to U$ is an isomorphism.*

Then there exists a U-admissible blowup $B' \to B$ such that U is scheme theoretically dense in B' and such that the strict transform X' of X maps isomorphically to an open subspace of B'.

Proof. This lemma is a generalization of Lemma 0GUW. As the composition of U-admissible blowups is U-admissible (Divisors on Spaces, Lemma 086C) we can proceed in stages. Pick a finite type quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_B$ with $|B| \setminus |U| = |V(\mathcal{I})|$. Replace B by the blowup of B in $\mathcal{I}$ and X by the strict transform of X. After this replacement $B \setminus U$ is the support of an effective Cartier divisor D (Divisors on Spaces, Lemma 085T). In particular U is scheme theoretically dense in B (Divisors on Spaces, Lemma 083S). Next, we do another U-admissible blowup to get to the situation where $X \to B$ is flat and of finite presentation, see Lemma 087E. Note that U is still scheme theoretically dense in B. Hence $X \to B$ is an open immersion by Lemma 0879. □